LoCRANE — Supplementary Material

Longevity-certified residual age network ensemble for transcriptomic aging clocks

Tim Shen

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This supplement accompanies the main text. All numbers are read from the frozen result tables released in `results/`; nothing is recomputed after evaluation. The locked protocol is `protocol_lock.json`.

1 Figure source map and band definitions

Every panel of the six result figures is generated from a frozen JSON/NPZ table.

Figure	Source table	Content
Fig. 1 (internal battle)	`x41_eq3_from_log.json`, `x20b_...json`, `x47_...json`, `x48_...json`	15-split lock, Holm for
Fig. 2 (ext1, GTEx blood)	`x45_ext_battle_v4.json`, `x15c_core3_kshot_v4.json`	zero-shot, k-shot, supp
Fig. 3 (ext2, GSE123696)	same as Fig. 2	cross-platform replicat
Fig. 4 (ablation)	`x42_ablation.json` + screen journal	component costs, six-a
Fig. 5 (attribution)	`x43_xai25.json` + attribution NPZ	25-seed gated axes
Fig. 6 (mining)	`MINING_V4.json` + citation audit	levers, modules, geome

Band definitions used in the per-band panels. Age bands are Y (≤ 50), M (51–89), L (≥ 90), following the cohort's verified longevity structure. In the external figures the age-band panels group each cohort's samples as out-of-support-young (below 50), in-support (50–89), and, for the microarray cohort, 90+; the support-coverage donuts print the exact arithmetic (32/64 and 27/66 out of the 50–89 training window). The internal per-band radial panel splits the working-age validation range into 50–62 / 63–75 / 76–89 thirds, with radial length proportional to excess MAE.

2 Per-split locked results

Mean absolute error per locked split (working-age validation, $n_{\text{test}} \approx 152$ per split). `eq3′` denotes the shipped hybrid; `base` the unregularized configuration; `pcc` the split-level Pearson correlation of `eq3′`. Paired comparison: 10/15 splits favor eq3′, two-sided Wilcoxon $p = 0.035$ (adoption gate, pre-registered).

split	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
eq3′	5.70	5.59	5.71	5.86	5.57	5.58	5.26	6.27	5.14	6.10	5.79	5.52	6.37	6.14	5.52
base	5.68	5.57	5.71	5.84	5.58	5.62	5.28	6.30	5.17	6.08	5.82	5.58	6.42	6.18	5.55

3 Full composition matrix

All fifteen equal-weight subsets of {deep, EN-40, EN-500, SVR-500} under the locked splits (`x48_composition.jso`
Headline eq3′ = 5.740; nearest rival deep+EN-40 at 5.747 (paired $p = 0.36$); single-member configurations significantly worse after Holm correction (deep-only 5.901, EN-40-only 5.839, SVR-500-only 5.898; Holm $p \leq 0.026$); best all-linear combination EN-40+SVR-500 at 5.806.

4 Notes on statistical practice and interpretation

1. **Deep-layer seed direction rate 0.56.** Claims are carried at the ensemble level (10/15, $p = 0.035$); the deep member alone is noisier than the conventional 0.80 bar.

2. **EN-500 is an ensemble member.** The 12/15, $p = 0.013$ comparison is a member-versus-frame comparison; all comparisons against non-member baselines sit at $p \leq 0.015$ (the maximum, the non-member XGBoost-500 row).

3. **External composition history.** The linear core's membership was identified with visibility into external few-shot results in an earlier round of the project. The neighboring composition {EN-40, SVR-500} tracks the core within 0.2 years everywhere, and the paper reports all combinations. A project-memo claim of bit-identical agreement between the core's k-shot table and the earlier-round table was found to be inaccurate (e.g., ext1 $k = 10$: 18.66 vs. 18.44 years); the frozen JSON tables are the source of record.

4. **Support coverage.** 32/64 (ext1) and 27/66 (ext2) samples lie outside the 50–89 training support; external claims are restricted to relative orderings under identical inputs.

5. **Attribution axis scope.** The FiLM per-gene attribution axis fails its pre-registered split-half gate ($\rho = 0.435 < 0.6$) and is reported as failed; FiLM's claims rest on its ablation cost and module-level dimorphism spectrum (split-half 0.70).

6. **PFKP grade correction.** PFKP's mortality association belongs to the source cohort study's supplementary screening table (HR 2.11, $p = 0.035$), not to its main signature; the paper carries the supplementary-screen grade. The locked replication here is directional (HR 1.20, $p = 0.17$).

7. **Geometry config-dependence.** The class-centroid velocity statistic is reported for the monotonicity-trained representation (mean 0.93, 21/25 seeds below 1); earlier non-regularized configurations showed expansion, so the statistic is tied to the shipped configuration.

8. **Unnominated genes.** ZNF518B and three further claims are graded unsupported by the literature audit and stand as falsifiable model-generated hypotheses.

9. **HVG pool overlap.** The 8,000-gene variance pool is an unsupervised ranking (no age labels enter it) computed on the working-age training samples of fold 0, which are reused by all fifteen locked splits. The pool's membership therefore shares sample provenance with every evaluation block. The ranking sensitivity to an alternative sample draw was not separately quantified; the pool was frozen at project start and held constant, so any such sensitivity acts on every arm identically. All arms, every baseline, and the external evaluations operate on the identical pool, and the 40- and 500-gene elastic-net feature sets are nested subsets of it.

10. **Adoption-p semantics.** The reported two-sided $p = 0.035$ is the monotonicity comparison on the eq3 layer (10 of 15 splits favor the adopted configuration); the deep-member layer shares the same p-value at 11 of 15 splits by coincidence. On the ten design-free splits (the last ten locked splits; the first four were design-exposed by the arm screen and one further split was conservatively excluded), the mean lead is +0.0299 years with two-sided $p = 0.0059$, so the adopted configuration's advantage survives the stricter exposure. The main text and the internal figure's panel K report the companion eleven-fresh-splits one-sided view (+0.028, one-sided $p = 0.002$; frozen recomputation, 10 of 11 splits in favor), and this two-sided ten-split value is the conservative statement of the same effect. p-value conventions: the classical-baseline and deep-suite comparisons of Table 2 and the composition matrix of Section 4.4 use one-sided paired Wilcoxon tests on the pre-declared direction (the challenger's higher error; frozen fields p_{less} / alternative = greater); the two adoption comparisons ($p = 0.035$; design-free $p = 0.0059$) and the ablation table are two-sided. Under the two-sided convention every family conclusion survives except one row: the EN-40-only composition row becomes non-significant (two-sided Holm $p \approx 0.051$), which the paper's one-sided convention does not report and these notes disclose.

11. **GSE123696 selection.** The second external cohort (Israeli PrimeView whole-blood microarray, $n = 66$, ages 23–96) was chosen at project start for population–platform independence from both the training cohort and the GTEx pilot. Its earlier-round evaluation predates the final lock, so a directional soft influence of that history on the final menu cannot be entirely excluded; only the post-lock evaluation is reported in the paper.

12. **Survival-null specification.** The survival probe (GSE65218, $n = 151$, 70 events) is sized by availability: power for the composite signature score reaches $\approx$ hazard ratios of 1.5 or larger only. Null readings are therefore *not-detectable at the available sample*, not evidence of absence.

13. **No unused external queue.** The project holds no formally registered unused third external dataset. The only candidate (GTEx muscle) was consumed in the first round, is same-source and same-country as the GTEx pilot, and does not qualify as an independent set. External claims are restricted to the two independent external cohorts; should a reviewer request a third queue, GSE65218 (used only for the survival probe) is disclosed here together with its usage history as a semi-independent supplement.

14. **GTEx pilot history.** The GTEx blood pilot (ext1, $n = 64$) entered the project as the first external evaluation and was re-evaluated on each pipeline round. The paper reports only the post-lock evaluation; no historical-round external numbers are repeated.

15. **p-value provenance.** Split-level p-values come from the repeated-split design: the fifteen 80/20 splits reuse one sample pool, so a paired Wilcoxon test on the split-level MAEs indicates replicate agreement under the locked design rather than independent-sample inference. The Table 2 and composition-matrix comparisons are one-sided on the pre-declared direction (the challenger's higher error); the adoption comparisons and the ablation table are two-sided. External p-values are paired bootstraps on external samples. No paper p-value is computed on data outside the evaluations.

16. **External p unadjusted.** Cell-level significance tests in the external table are not multiplicity-adjusted across cells; they are reported per comparison, on the conservative per-ensemble

maximum basis (the maximum over the four per-baseline paired bootstrap comparisons; ext2: $p \leq 0.033$).

17. **Collapse control provenance.** The 10.714-year collapse control is computed under the identical fifteen locked splits and 80/20 protocol, but in the pipeline generation preceding the final result tables, where the elastic-net baseline was handed the extreme-age labels as ordinary regression targets. The final tables do not include that control arm; the value is a cross-generation structural control, not a fresh re-run on the shipped configuration.

18. **Survival p nominal.** In-band gene-level survival p-values (GAL3ST4 $p = 0.007$; PLEKHG4 $p = 0.057$; PDZK1IP1 $p = 0.095$; PFKP $p = 0.17$; NT5E $p = 0.85$; BCL2 $p = 0.116$; CPNE5 $p = 0.256$; ten further genes $p \geq 0.26$) are nominal and unadjusted across the 17 in-band gene-level tests (frozen table `x31_sensitivity`). None survives a Bonferroni correction at $\alpha = 0.05/17 \approx 0.003$ (the smallest, GAL3ST4 $p = 0.007$, does not), and the paper presents them as directional, corroborating evidence, not as a survival signature.

19. **Ablation p uncorrected.** Paired p-values in the ablation table are uncorrected (descriptive family); the Holm correction is applied only to the two registered comparison families.

20. **Per-ensemble-max p basis.** The p-basis stated for the external table is the maximum over the four per-baseline comparisons for each ensemble (conservative); cell-level p-values are not reported in the paper.

21. **Holm variant.** The Holm step-down correction is computed within each registered comparison family (the classical-baseline family; the member-versus-frame family), not jointly across all pairwise comparisons. The family-corrected p-values are the ones reported in the internal table.

22. **Figure-level statistics.** Four figure-level quantities (the $\rho = 0.89$ agreement of the M2L–M2Y attribution axes; the 20-of-30 gene overlap across the two configurations; the 20-gene layer-participation count of the linear core (Fig. 6 panel K); and the neighboring-composition label {EN-40, SVR-500}) are read from the `x43`/`x45` companion NPZ tables and project memos rather than from a runs_v4 JSON field; their values agree with the frozen JSONs that do carry the corresponding per-value data (neighboring-composition gaps ≤ 0.19 years across all six cells).

5 Survival probe details (Vitality 90+, GSE65218)

Cohort: 151 Finnish nonagenarians, Illumina HumanHT-12 v4, 70 deaths over the follow-up window (Jylhävä et al., BMC Medical Genomics 7:54, 2014). Pre-registered hypotheses, fully locked consensus weights: (H1) signature-score Cox model adjusted for frailty and sex — HR 1.14 (95% CI 0.89–1.45, $p = 0.31$), confirmed by the frozen permutation check (`x31_sensitivity`: `perm_p_frac`= 0.3125, the observed p at the null quantile of the permutation distribution); (H2) tertile Kaplan–Meier separation — log-rank $p = 0.47$; (H3) per-gene directional agreement — 7/10 testable genes agree with the model (70%). Nested cross-validated recalibration of the weights reaches C-index 0.540 against 0.585 for covariates alone ($n = 151$; 70 events power only HR ≥ 1.5). Gene-level associations inside the 90+ band (frozen tables `x30`/`x31`; 17-gene test family): GAL3ST4 HR 1.39 (1.09–1.76, $p = 0.007$); PDZK1IP1 HR 1.21 ($p = 0.095$); PFKP HR 1.20 ($p = 0.17$, directional replication of the supplementary screen); NT5E HR 1.03 ($p = 0.85$, null). Reading: the

longevity signature answers band membership, not survival within the band; the two questions decouple.

6 Demo reproduction

`python code/run_demo.py` runs the full recipe (anchor, FiLM, band auxiliary, monotonicity, hybrid blend) on the released real slice in about two minutes on CPU. Demo-scale numbers verify the pipeline only; the paper's numbers require the full cohort (see the Data section of the repository README).